

HOSPITAL MANAGEMENT SYSTEM

HMS Core Web Application Documentation

Product documentation for the React and Vite based hospital management demo app. This guide covers architecture, login flow, navigation, module capabilities, storage behavior, and screen references captured from the current build.

Application

HMS Core v1.0.0
Premium

Generated

17 March 2026

Source

src/, public/,
docs/screenshots/

PURPOSE

Provide a readable handoff artifact for demos, reviews, and onboarding without needing to inspect the codebase directly.

- | | |
|---------------------------------|-------------------------------|
| 1. Overview and stack | 2. Login and persistence |
| 3. Operational module summaries | 4. Storage and workflow notes |
| 5. Screenshot gallery | 6. Final deliverables |

Application Overview

HMS Core is a single-page web app that simulates core hospital workflows across patient administration, scheduling, records, billing, pharmacy, diagnostics, staffing, logistics, and blood inventory.

Technology Stack

- Frontend: React 19 with React Router 7
- Bundler: Vite 7
- UI foundation: Bootstrap 5 with custom CSS
- State management: React context and local component state
- Persistence: Browser `localStorage` for session and demo data

Core UX Pattern

- Protected routes redirect unauthenticated users to `/login`
- Sidebar navigation exposes 12 operational modules
- Navbar provides search feedback and light or dark theme toggle
- Each primary module exposes seeded data and an action modal or shortcut

Navigation Model

The application uses a single shell layout with a persistent sidebar and a fixed top navbar. Each module is loaded through React Router and most pages are lazy-loaded.

Data Model Summary

Shared operational data is held in one context provider, while a few dashboard and logistics style pages render from local static arrays for demo presentation.

12

Routed modules

7

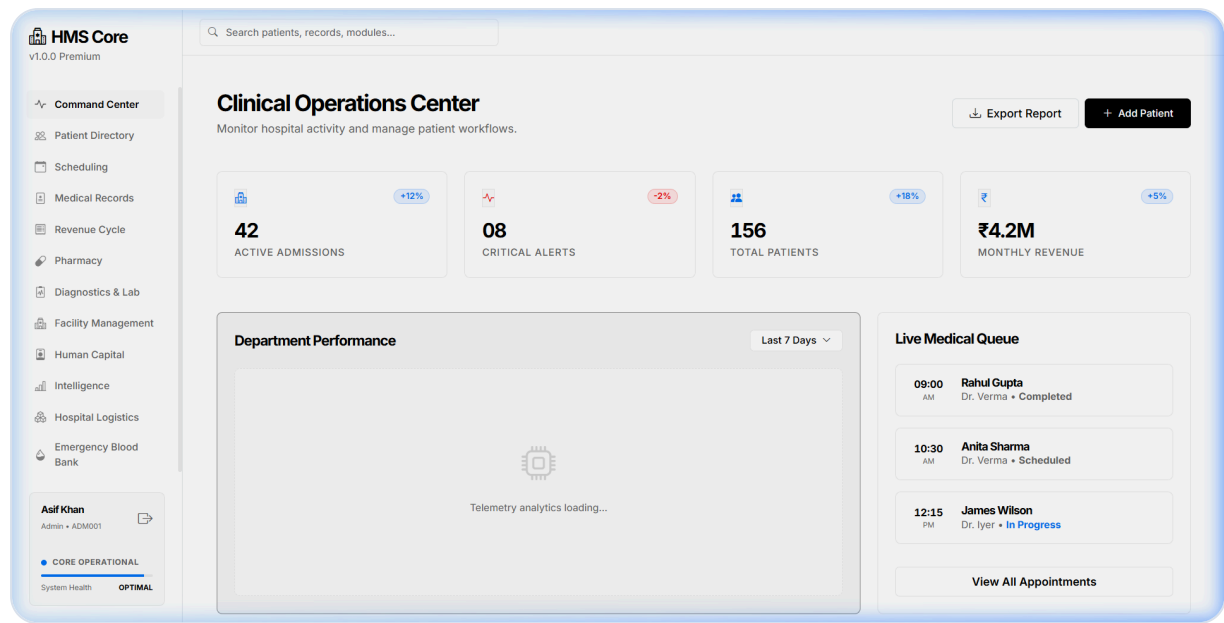
Persisted data collections

1

Context provider

13

Reference screenshots



Dashboard view showing the command center, KPI cards, department performance area, and live appointment queue.

Login, Access, and Persistence

Authentication is demo-only and enforced entirely on the client. Successful login stores a mock user object in `localStorage`, and protected routes read that session state on load.

Demo Credentials

- Username: asifkhan
- Password: asifkhan@349
- Authenticated role: Admin
- User identifier: ADM001

Persisted Browser Keys

- hms_user
- hms_theme
- hms_patients, hms_appointments, hms_records
- hms_invoices, hms_medicines, hms_tests, hms_staff

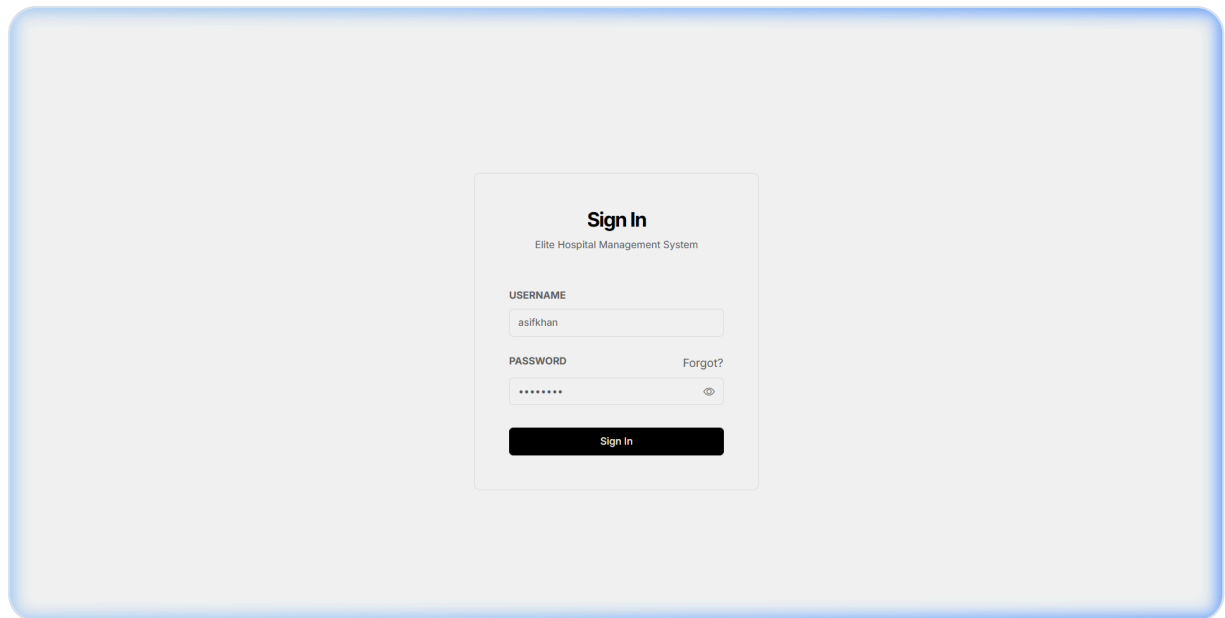
User opens a protected route and is redirected to the login screen if no saved session exists.

Successful login writes a mock admin user object to localStorage.

The selected theme is re-applied on every reload through a root data attribute.

Operational records created in the app remain available until browser storage is cleared.

Area	Implemented Behavior	Documentation Note
Login	Validates one hard-coded username and password pair.	Suitable for demos only, not production security.
Route protection	Unauthenticated requests are redirected to `/login` with the original route saved.	Provides a smooth post-login return flow.
Theme persistence	Light and dark theme choice is stored and re-applied on reload.	Useful for user preference continuity.
Operational data	User-created demo records remain in the browser until storage is cleared.	No backend sync, audit trail, or multi-user consistency.



Login screen used to enter the seeded administrator credentials and access protected routes.

Module Guide I

The first group of modules covers operational intake, scheduling, clinical records, and revenue handling.

Command Center

Landing dashboard with KPI cards, export shortcut, and a live appointment queue.

- Displays static KPI values for admissions, alerts, patients, and revenue.
- Uses live appointment data from context for the queue card.
- Navigates directly to patient registration and appointment listing.

Patient Directory

Registry for patient demographics, blood groups, and admission status.

- Lists seeded and user-added patient profiles.
- Modal supports registration with age, gender, blood group, and status.
- "Access EHR" links the workflow to the EMR module.

Scheduling

Daily queue for appointments with doctor, slot, visit type, and status.

- Bookings are created from existing patient records.
- New appointments default to the Scheduled state.
- Calendar and edit actions currently trigger toast feedback only.

Medical Records

Electronic records table for diagnosis, clinician, and prescription tracking.

- Creates new entries with generated clinical reference IDs.
- Stores diagnosis and prescription data in client-side state.
- Search and report actions are present as UI placeholders.

Revenue Cycle

Invoice management with patient mapping, amount entry, and payment status.

- New invoices start in a `Pending` state with an unset payment method.
- Amounts are formatted as rupee values before saving.
- Print, PDF, export, and sync buttons currently simulate actions.

Module Guide II

The second group focuses on pharmacy, diagnostics, facility management, people operations, and analytics.

Pharmacy

Medication stock tracking with batch number, quantity, expiry, and stock status.

- New stock can be added through a modal form.
- Entries with more than 20 units are visually treated as healthy stock.
- Dispense and stock alert functions are UI-only demo actions.

Diagnostics and Lab

Orders and tracks pathology or imaging requests for registered patients.

- Lab orders are linked to a patient and doctor.
- New tests default to Pending Approval.
- Processing and publishing actions are present for demonstration.

Facility Management

Visual bed board grouped into ICU, general ward, and private rooms.

- Shows availability, occupied beds, cleaning, and maintenance states.
- Operates from local static arrays, not shared context data.
- Bed click handling currently logs status rather than opening a detail view.

Human Capital

Staff directory for role, department, shift, and duty status.

- Supports onboarding new staff through a modal workflow.
- Directory data is persisted in browser storage.
- File edit and messaging buttons are simulated actions.

Intelligence

Analytics catalog with pre-defined reports and an advisory insight block.

- Shows report categories, last generated timestamp, and sync actions.
- Includes a Gen-AI style advisory message for staffing scale-up.
- Does not yet render real charts or downloadable analytics payloads.

Module Guide III

The remaining modules address logistics and emergency supply visibility. Both are useful as executive snapshots, but they currently operate from static seeded arrays.

Hospital Logistics

Supply table for high-use hospital consumables and PPE.

- Shows item ID, category, current stock, consumption, and status.
- Actions include audit generation, supply request, and per-item reorder.
- Dataset is memoized inside the page rather than shared across modules.

Emergency Blood Bank

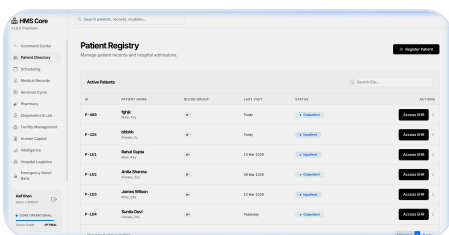
Blood group stock cards and a compact activity log for donations and requests.

- Displays stable, low, and critical inventory states.
- Activity feed includes donors, hospitals, and unit movement.
- Donor intake and request registry actions are placeholders.

Module	Primary Data Source	User Data Saved
Patients, Appointments, EMR, Billing, Pharmacy, Lab, Staff	Shared `AppContext` state	Yes, in browser `localStorage`
Dashboard	Mixed: static KPI data plus live appointments	No direct edits from the page
Beds, Inventory, Blood Bank, Reports	Static in-page arrays	No persistent edits

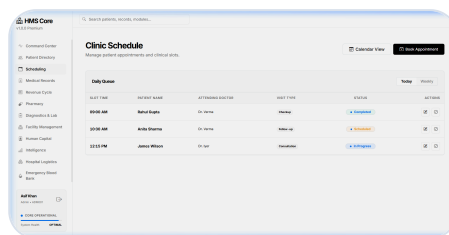
Screen Reference Gallery

The following screens are included as visual references for stakeholders and onboarding.



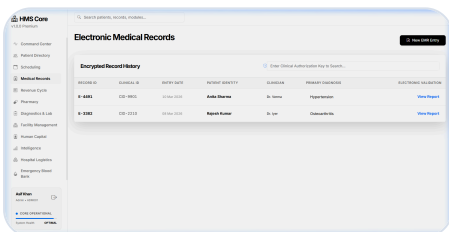
Patient Directory

Registration workflow, active profiles table, and EHR handoff action.



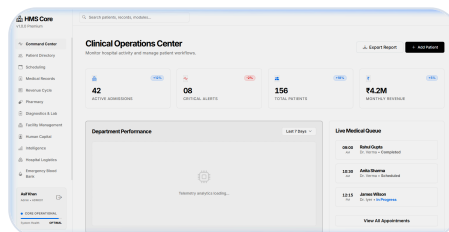
Scheduling

Appointment slots with status tags and booking actions.



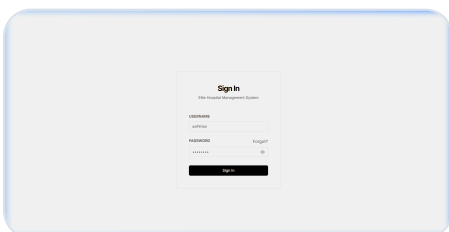
Medical Records

Clinical history ledger with encrypted record styling and entry controls.



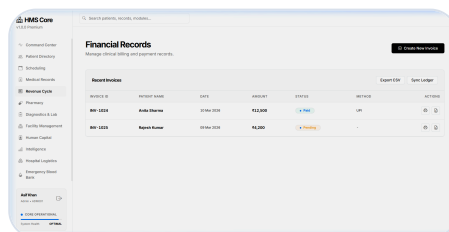
Command Center

Top-level operations summary with queue visibility and quick actions.



Login

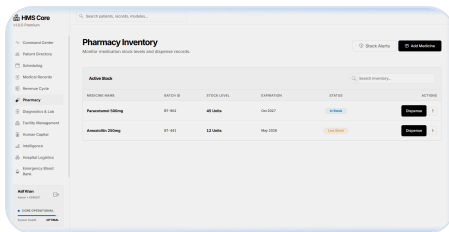
Entry point for protected routes using the seeded admin account.



Revenue Cycle

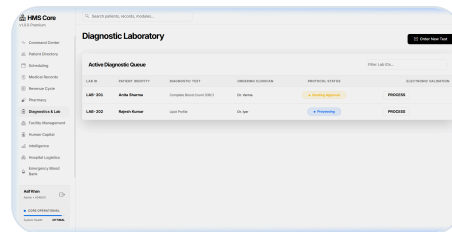
Invoice listing with export, sync, print, and PDF affordances.

Additional Screens and Delivery Notes



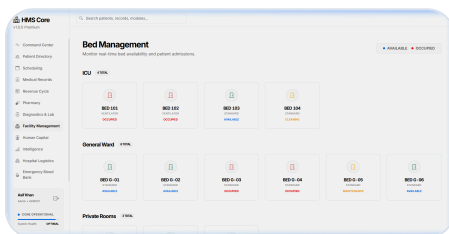
Pharmacy

Medication stock management with add and dispense controls.



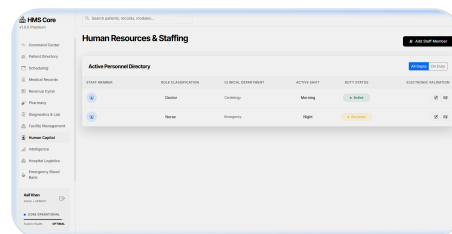
Diagnostics and Lab

Diagnostic queue with processing and validation actions.



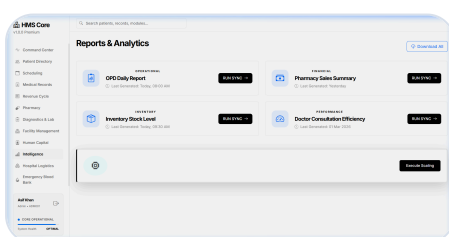
Facility Management

Visual map of ICU, ward, and room availability.



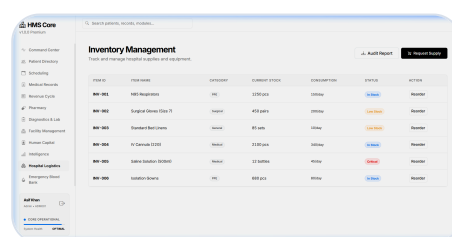
Human Capital

Staff directory with onboarding flow and shift tracking.



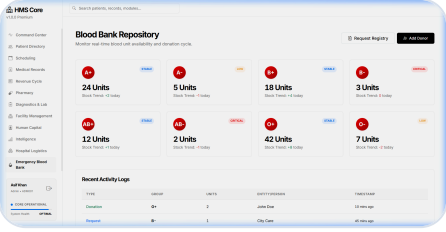
Intelligence

Operational report catalog and advisory recommendation panel.



Hospital Logistics

Supply monitoring table with reorder calls to action.



Emergency Blood Bank

Blood stock dashboard with activity log.

Deliverables generated for this repository: docs/web-app-documentation.html, docs/documentation.css, and docs/HMS-Web-App-Documentation.pdf.